8 The isotope uranium-235 is represented by

$$^{235}_{92}U.$$

(a) State what the numbers 92 and 235 represent in this symbol.

92 is .....

235 is .....

[2]

(b) Uranium-235 is a fuel used in nuclear reactors.

(i) State the process by which energy is released from uranium-235 in a nuclear reactor.

..... [1]

(ii) A nuclide equation for this process is

$$^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{140}_{54}Xe + {}^{94}_{38}Sr + 2\,{}^{1}_{0}n.$$

Describe the mass and energy changes that take place during this process in a nuclear reactor.

.....

.....

..... [2]

(c) **(i)** Describe how thermal energy from nuclear reactions is used to generate electricity in a power station.

.....

.....

.....

..... [3]

(ii) State **one** advantage and **one** disadvantage of using nuclear fuels in a power station instead of using fossil fuels.

advantage .....

.....

disadvantage .....

.....

[2]

[Total: 10]

© UCLES 2023